Portfolio Allocation Strategies under Maximal Drawdown Risk

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August 28, 2025

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Chapter 1

Introduction

Portfolio management has evolved through centuries of financial thought, blending theory, empirical evidence, and investor psychology. One of its key concerns is critical task of distributing capital across asset classes to align with an investor's objectives and risk tolerance. Since the seminal work of Markowitz (1952) formalised modern portfolio theory, the field has evolved to include a diverse range of methodologies, from passive buy-and-hold approaches to dynamic strategies grounded in behavioural finance. This thesis aims to contribute to this discourse by conducting a rigorous empirical evaluation and comparison of five distinct portfolio allocation strategies across sharply divergent market regimes.

The primary objective is to assess the performance and robustness of fundamental approaches—including Buy and Hold (BAH), Constant Rebalanced Portfolio (CRP), Momentum, and Contrarian strategies—alongside a hybrid Aggregated strategy that dynamically combines them. This investigation is designed to move beyond a singular market condition, testing these strategies in both a high-growth, bullish environment and a stressful, bearish environment comprising underperforming assets. This dual framework will benchmark performance under ideal conditions while simultaneously stress-testing each strategy's defensive capabilities and inherent vulnerabilities during prolonged adversity.

Each approach reflects a different philosophy about how markets behave: whether they are efficient, prone to overreaction, or characterized by mean reversion. This thesis explores the origins, motivations, and academic contributions surrounding each strategy, situating them in both theoretical and practical contexts.

A further key aim is to employ a performance evaluation framework specifically suited to the study's focus on capital preservation and behavioural realism. Maximal Drawdown (MDD) is chosen as a suitable metric to capture the specific dimensions of risk under investigation here. MDD, which quantifies the worst peak-to-trough loss, provides a path-dependent measure of capital erosion that directly aligns with investor psychology during periods of significant stress and the practical mandates of institutional investors focused on limiting severe losses. Therefore, its use is not intended to question the utility of other established metrics but to provide a complementary, clinically relevant perspective on downside risk that is particularly informative for the stress-testing aims of this study.

Ultimately, the overarching goal is to explore the concept of strategic diversification. This research seeks to determine whether resilience is better achieved not merely by diversifying across assets, but by diversifying across the underlying decision-making paradigms themselves. By constructing a meta-strategy that dynamically allocates to a pool of sub-strategies, the thesis aims to investigate a path toward a more adaptive and robust portfolio management framework. Through this empirical investigation, this work

seeks to provide practical insights into constructing portfolios that can pursue long-term growth while mitigating the severity of capital erosion, a balance of paramount importance to the real-world concerns of investors.

Chapter 2

Literature Review

Portfolio allocation remains a foundational challenge in financial economics and investment management, both at an academic and a practical level. The origins of portfolio allocation theory can be traced back to the early twentieth century, but it was the formalization by Harry Markowitz (1952) that initiated a revolution, introducing the concept of diversification and the efficient frontier. At its core, the allocation problem asks how to distribute capital—often across a wide array of asset classes, such as stocks, bonds, commodities, and alternatives—in a manner that aligns with an investor's goals, risk tolerance, and time horizon. The complexity of this task has steadily increased in the face of globalized financial markets, rapid innovation in investment products, and the emergence of new risks and opportunities. In academic research, portfolio allocation is situated at the intersection of economics, probability theory, and behavioral finance, reflecting the rich interplay between quantitative modeling and the real-world decisions faced by practitioners. The policy implications of sound portfolio allocation resonate widely, influencing not only professional fund managers and institutional investors, but also individuals saving for retirement or other long-term financial targets. Ultimately, the capital allocation decision shapes the trajectory of wealth accumulation, risk exposure, and resilience to market disruptions, making it a central problem that has inspired decades of research and debate. Since the foundational work by Markowitz (1952), a rich field of portfolio theory and quantitative strategy design has evolved, giving rise to multiple allocation methodologies, each built on distinct assumptions about market behavior and investor priorities.

A traditional approach, known as Buy and Hold (BAH), is rooted in the Efficient Market Hypothesis (EMH) and the belief in long-term economic growth as reflected in asset prices (Fama, 1970). In this framework, investors allocate their capital according to initial target weights and abstain from any further trading, regardless of evolving price dynamics or new information. BAH's appeal lies in its simplicity, low transaction costs, and the ability to harness the compounding effects of asset appreciation over time (Bogle, 1999). Nevertheless, this strategy is not without its limitations: BAH leaves investors vulnerable to severe drawdowns during market downturns or in stagnating sectors, since it does not adapt to adverse trends or shifts in the macroeconomic environment. The success of BAH thus depends heavily on the initial asset selection and the persistence of long-term growth trends.

Cases where BAH has proven particularly effective include the postwar expansion in U.S. equities, where investors who maintained broad exposure from the 1950s through the late 1990s experienced substantial real wealth accumulation (Siegel, 1998). The outperformance of low-cost S&P 500 index funds relative to the vast majority of actively

managed funds from 1970 to 2010 further highlights BAH's value in minimizing costs and delivering compounding returns (French, 2008; Malkiel, 1995). Another prominent example came after the 2008 Global Financial Crisis, when investors who held diversified stock portfolios throughout the downturn and recovery recouped losses and realized significant long-term gains by 2013 (Bogle, 2012).

However, there are also pronounced instances in which Buy and Hold strategies have underperformed or exposed investors to severe losses. The Japanese stock market's experience since the late 1980s offers a striking example: investors who purchased and held the Nikkei 225 at its 1989 peak faced decades of drawdowns and incomplete recovery (Campbell, Shiller and Schoenholtz, 2001). Sector-specific BAH approaches can be equally hazardous, as seen in the long-term decline of media and telecommunications stocks after the dot-com bubble, where many constituent firms failed to recover or even exited the market (Ofek and Richardson, 2003). Moreover, individual stocks such as Kodak or General Motors, once blue-chip staples, provide cautionary cases where BAH led to permanent capital loss as industry disruptions rendered business models obsolete (Friedman, 2011).

Empirical evidence has demonstrated that while BAH portfolios—such as broad-market index funds—have outperformed a majority of actively managed funds over multiple decades, they can also experience prolonged periods of underperformance and deep losses during market crises (Siegel, 1998; French, 2008).

Responding to the inflexibility of BAH, more active strategies such as Constantly Rebalanced Portfolios (CRP) have gained traction in both theory and practice. CRP strategies, formalized by Cover (1991), maintain constant target weights through periodic rebalancing, buying and selling assets at fixed intervals to restore the original allocation. This process exploits mean-reverting behavior in asset prices, allowing investors to sell assets that have risen relative to their targets and buy those that have fallen—effectively "harvesting volatility." The mathematical underpinnings of these strategies are captured within the framework of stochastic portfolio theory (Fernholz and Shay, 1982). Empirical studies have shown that CRP can outperform BAH, particularly in markets characterized by frequent oscillations and reversals, provided transaction costs remain low (DeMiguel, Garlappi and Uppal, 2009).

CRP and constant-mix strategies have found strong success in several important scenarios. During the 1973–1986 period, CRP portfolios applied to the U.S. equity-bond markets generated superior risk-adjusted returns versus BAH, as rebalancing took advantage of market volatility and mean reversion (Perold and Sharpe, 1988). Similarly, CRP strategies delivered higher Sharpe ratios and lower drawdowns in mixed-asset portfolios during the turbulent 1970s inflation era, when commodity and equity returns often moved in opposite cycles (Ilmanen, 2011). In the context of international markets, the "universal portfolio" variant of CRP outperformed non-rebalancing allocations in diversified equity baskets exposed to Asian, European, and U.S. stocks over multi-decade time frames (Cover and Ordentlich, 1996).

However, the limitations of CRP are evident under certain persistent trends or high transaction cost environments. CRP strategies performed poorly in the U.S. stock market throughout the 1995–1999 technology-led bull market, as continual rebalancing away from outperforming tech equities led to missed long-term gains (French, 2008). During the roaring bull run from 2009 to 2021, when U.S. equity outperformance was unusually consistent, constant-mix approaches underperformed BAH and momentum, as they systematically cut exposure to winning assets (Asness, Frazzini and Pedersen, 2015).

Furthermore, research shows that in real-world trading with even modest transaction costs, the potential benefits of CRP strategies are often offset or reversed; DeMiguel, Garlappi and Uppal (2009) found that accounting for trading frictions could fully erode the advantages of high-frequency rebalancing in both simulated and empirical backtests.

Momentum-based strategies challenge the predictions of the EMH by betting on the empirical regularity that assets with strong recent performance tend to continue outperforming in the near future. The foundational work by Jegadeesh and Titman (1993) showed that sorting assets on past returns and investing in recent winners yields significant positive excess returns—a finding documented across a diverse set of asset classes and geographies. The theoretical literature attributes momentum profits to various behavioral factors, such as investor herding, slow information diffusion, and the underreaction to news (Hong and Stein, 1999).

Momentum's success is supported by multiple high-profile empirical findings. First, Jegadeesh and Titman (1993, 2001) demonstrated pronounced positive returns to cross-sectional equity momentum strategies in U.S. stocks during the 1965–1989 period, a result that has held across international equity markets as well (Rouwenhorst, 1998). Second, Moskowitz, Ooi and Pedersen (2012) reported consistent outperformance of trend-following momentum in diverse global markets—including equities, commodities, currencies, and bonds—over a century of data. Third, Barroso and Santa-Clara (2015) found that risk-managed equity momentum strategies delivered attractive risk-adjusted returns and lower drawdowns during volatile market environments.

Despite these successes, momentum is vulnerable to sharp reversals. Daniel and Moskowitz (2016) documented that momentum suffered dramatic "crashes" during strong market recoveries, such as the 2009 post-crisis rebound, where winners sharply lagged behind laggards. Fama and French (1996) noted that momentum approaches can yield sustained losses during certain market turnarounds, including the Asian Financial Crisis of 1997–1998, when reversal dynamics dominated. Also, momentum strategies employed in emerging markets, such as Brazil and Turkey during episodes of rapid regulatory change and market intervention, have experienced periods of significant underperformance or even negative returns (Ho and Cheung, 2016).

Nevertheless, momentum investing is not without risk: portfolios constructed on momentum signals can be subject to sharp reversals—so-called "momentum crashes"—especially when market trends suddenly dissipate or reverse. Frequent rebalancing also increases turnover and associated costs, sometimes offsetting the gross returns from momentum effects.

In contrast, contrarian or anti-momentum strategies intentionally target recent losers based on the premise of mean-reverting price behavior. Classic studies, such as De Bondt and Thaler (1985), empirically documented that securities with poor past performance tend to outperform former winners over longer horizons, attributing these return reversals to market overreaction and subsequent correction. The appeal of contrarian investing lies in its ability to capitalize on sentiment-driven overshooting and subsequent realignment with fundamental value.

Examples of successful contrarian strategies include the long-term winner-loser reversals observed in U.S. equities, where the portfolios buying the worst performers of the prior three to five years outperformed portfolios of prior winners, as documented by De Bondt and Thaler (1985, 1987). International markets have echoed these findings, with Japan and the U.K. displaying similar return patterns, especially during periods following dramatic market busts (Richards, 1997). Another instance of contrarian outperformance was noted

following the bursting of the dot-com bubble, where previously distressed technology stocks experienced significant rebounds (Chan, Jegadeesh and Lakonishok, 1996).

Nonetheless, contrarian strategies have faced marked failures in several contexts. One risk manifests in the persistence of structural decline or disruption, such as value traps—where companies appearing as bargains continue to underperform or fail. Recent decades have seen such effects in declining sectors like coal mining or print media, where mean reversion failed to materialize and losses deepened for contrarian investors (Lakonishok, Shleifer and Vishny, 1994). The energy sector's protracted downturn from 2014–2020 further illustrates this hazard, as early contrarian bets on oil and gas stocks led to severe drawdowns before conditions stabilized (Asness, Frazzini and Pedersen, 2015). Empirical backtests also reveal that contrarian approaches can suffer during persistent bull trends, notably in emerging markets such as China during the 2005–2007 rally, when previous losers continued to lag as broad uptrends persisted (Chui, Titman and Wei, 2010).

Given that contrarian strategies can endure extended drawdown periods and may require high portfolio turnover, successful implementation often demands patience and robust risk management.

Recent advances in portfolio theory have advocated for aggregating and combining distinct strategies to achieve more robust risk-adjusted returns. Cover's (1991) Universal Portfolios theorem presents a formal method of capital allocation across a pool of strategies or "expert portfolios," such that the aggregate portfolio asymptotically matches the performance of the best-in-hindsight constituent strategy (Ordentlich and Cover, 1998). Empirical work has confirmed the benefits of aggregation—by dynamically allocating across momentum, contrarian, CRP, and BAH methodologies, investors can often reduce maximum drawdowns and improve resilience against regime shifts and market anomalies.

Several case studies and large-scale analyses support the power of strategy aggregation. First, the application of Cover's Universal Portfolio algorithm to historical U.S. stock data has been shown to match or exceed the performance of the best fixed-mix strategy chosen in hindsight, even in environments characterized by dramatic regime shifts (Cover and Ordentlich, 1996). Second, multi-strategy funds such as AQR's Absolute Return Fund and Bridgewater's All Weather portfolio have successfully combined value, momentum, and risk-parity approaches to produce more stable returns with lower drawdowns in the 2000s and 2010s (Ilmanen, 2011; Asness, Frazzini and Pedersen, 2015). Third, international evidence indicates that dynamic allocation across major equity and alternative risk premia has improved risk-adjusted performance relative to static or single-strategy approaches (Fung and Hsieh, 2004).

Despite these successes, aggregation also presents challenges and notable periods of disappointing performance. For example, in the wake of the 2008 financial crisis, many risk-parity and multi-strategy funds struggled with high correlations across asset classes, which undermined the benefits of diversification and strategy mixing (Harvey, Liechty, Liechty and Muller, 2013). Similarly, during the sharp COVID-19 market shock of March 2020, diversified multi-asset and quant macro funds exhibited larger-than-expected drawdowns due to simultaneous losses in equities, bonds, and trend-following commodities (Lopez de Prado, 2020). Additionally, empirical research cautions that backtested universal strategies often rely on frictionless assumptions and, when applied net of trading costs or in more illiquid markets, may suffer significant performance decay or even underperform simpler alternatives (van Vliet and de Koning, 2016).

While traditional portfolio analysis was often dominated by the mean-variance paradigm,

modern research increasingly acknowledges the importance of path-dependent risk measures, such as maximal drawdown, in capturing investors' true risk exposures. Maximal drawdown quantifies the worst peak-to-trough loss experienced by a portfolio over a specified time frame, offering a more investor-centric and actionable metric than symmetrical volatility-based measures (Magdon-Ismail, Atiya, Pratap and Abu-Mostafa, 2004; Chekhlov, Uryasev and Zabarankin, 2005). This focus is particularly relevant in contemporary markets, where shocks, structural breaks, and behavioral dynamics can drive extreme outcomes, challenging both passive and active asset allocation approaches alike.

Chapter 3

Methodology

This chapter outlines the methodological framework used to evaluate the performance of different portfolio allocation strategies. The analysis is structured around two main components: a theoretical foundation, which specifies the mathematical representation of portfolio growth, rebalancing rules, and transaction costs; and an empirical implementation, which translates these concepts into computational models reproducible under a variety of contexts. The central goal is to design, simulate, and assess the effectiveness of several allocation strategies under realistic market conditions, accounting for trading frictions and considering both strong-performing and distressed asset classes.

3.1 Theoretical Framework of Portfolio Allocation

The investment process is modeled as a discrete sequence unfolding over T trading periods. Each portfolio consists of $m \geq 1$ assets, whose closing prices at time t are denoted by $X_t^1, X_t^2, \ldots, X_t^m \geq 0$, and one risk-free cash asset, assumed to yield zero interest for simplicity. The net return of asset i from period $t-1$ to t is defined as:

$$R_t^i = \frac{X_t^i - X_{t-1}^i}{X_{t-1}^i}. \tag{3.1}$$

The corresponding growth factor for the asset over the same period is $(1 + R_t^i) = X_t^i / X_{t-1}^i$, representing a profit if greater than or equal to one, and a loss otherwise. The logarithmic return, $\log(1 + R_t^i)$, provides a convenient approximation for compounding across multiple periods. The cumulative growth of a single asset from time 0 to T is obtained by multiplying the growth factors of each period:

$$\frac{X_T^i}{X_0^i} = \prod_{t=1}^{T} (1 + R_t^i). \tag{3.2}$$

An allocation strategy is defined by a vector $\boldsymbol{\theta}_t = (\theta_t^1, \theta_t^2, \ldots, \theta_t^m)$ that specifies the proportion of the total portfolio value invested in each asset at the beginning of period t. The strategy is restricted to long-only positions, so all weights satisfy $(\theta_t^i \geq 0)$. Leverage is not allowed, implying $(\sum_{i=1}^m \theta_t^i \leq 1)$. The remaining proportion, $1 - \sum_{i=1}^m \theta_t^i$, is allocated to cash. A critical assumption is that the strategy $\boldsymbol{\theta}_t$ can depend only on information available up to time $t-1$, ensuring no look-ahead bias (i.e., the artificial use of future information in portfolio decisions, which would otherwise lead to overstated performance).

Given an initial capital V_0, the evolution of the portfolio value is governed by the recursive equation:

$$V_t = V_{t-1}\left(1 - \sum_{i=1}^{m} \theta_{t-1}^i + \sum_{i=1}^{m} \theta_{t-1}^i \frac{X_t^i}{X_{t-1}^i}\right) = V_{t-1}\left(1 + \sum_{i=1}^{m} \theta_{t-1}^i R_t^i\right). \tag{3.3}$$

The log-value of the portfolio at terminal time T is therefore given by the sum of the log-periodic returns:

$$\ln V_T(\boldsymbol{\theta}) = \sum_{t=1}^{T} \ln\left(1 + \sum_{i=1}^{m} \theta_{t-1}^i R_t^i\right). \tag{3.4}$$

3.2 Investment Strategies

The Buy-and-Hold (BAH) and Constant Rebalanced Portfolio (CRP) are foundational strategies providing two simple but contrasting benchmarks. BAH is passive: once the initial allocation is set, no further trading takes place. Portfolio weights change automatically with asset price movements, which can lead to concentrated positions in assets that perform well. This approach avoids transaction costs but does not control diversification over time. CRP, in contrast, is active: portfolio weights are reset to fixed targets at each rebalancing step. This requires selling assets that have gained value and buying those that have lost value, which keeps the portfolio diversified but introduces continuous trading and transaction costs. Taken together, BAH and CRP highlight two extremes in portfolio management—one that fully accepts drift with the market and another that enforces constant balance—making them useful reference points for assessing more advanced strategies.

3.2.1 Buy-and-Hold Strategy

The first benchmark strategy is the **Buy-and-Hold (BAH)** portfolio. It is defined by constant initial weights θ_0^i, which determine the fraction of capital invested in each asset at time 0. Once established, these allocations remain unchanged throughout the investment horizon.

Its terminal value is simply:

$$V_T = \left(1 - \sum_{i=1}^{m} \theta_0^i\right) + \sum_{i=1}^{m} \theta_0^i \frac{X_T^i}{X_0^i}. \tag{3.5}$$

In practice, while the quantity of each asset purchased remains constant, the proportion of wealth in each asset, denoted by θ_t^i, changes over time. This drift is caused by the relative price movements of the assets. For example, if one asset grows faster than the others, it will gradually represent a larger share of the portfolio.

For a uniform BAH portfolio with equal initial allocation ($\theta_0^i = 1/m$), the dynamic weight at time $t-1$ is expressed as:

$$\theta_{t-1}^i = \frac{X_{t-1}^i / X_0^i}{\sum_{j=1}^{m} X_{t-1}^j / X_0^j}. \tag{3.6}$$

Thus, the Buy-and-Hold strategy reflects a passive approach where weights evolve naturally with asset price dynamics, without any further trading after the initial investment.

3.2.2 Constantly Rebalanced Strategy

The second benchmark is the **Constant Rebalanced Portfolio (CRP)**. Unlike BAH, this strategy maintains fixed target weights, $\theta_t^i = 1/m$ for an equally weighted CRP. After each period, the portfolio is rebalanced to restore the target proportions.

This requires selling a fraction of assets that have increased in value and reinvesting in those that have fallen behind. As a result, CRP systematically transfers capital from recent winners to recent losers.

The portfolio value evolves according to:

$$V_t = V_{t-1}\left(\sum_{i=1}^{m}\frac{1}{m}\frac{X_t^i}{X_{t-1}^i}\right) = \prod_{s=1}^{t}\left(1+\sum_{i=1}^{m}\frac{1}{m}R_s^i\right). \tag{3.7}$$

This mechanism enforces mean-reversion across assets and ensures that portfolio weights remain stable through time. In contrast to BAH, which drifts with market trends, CRP demands continuous rebalancing and trading activity.

3.2.3 Specification of Dynamic Strategies

Beyond the static BAH and CRP benchmarks, dynamic strategies are formulated based on price momentum. For these strategies, a **lookback window** of $L = 52$ weeks (one year) is employed, consistent with empirical literature that identifies the 6-12 month horizon as optimal for momentum persistence.

For each asset i, the rolling maximum and minimum prices over the lookback window $[t-L, t]$ are calculated:

$$(X^*)^i(t-L,t) = \max_{r\in[t-L,t]} X^i(r), \quad (X_*)^i(t-L,t) = \min_{r\in[t-L,t]} X^i(r). \tag{3.8}$$

Two opposing signals are then generated. The **momentum strategy** for asset i is predicated on the assumption that assets near their highs will continue to outperform. Its weight is proportional to the proximity to the rolling maximum:

$$\theta_{\text{mom}}^{(i)}(t) = \left(0,\ldots,\underbrace{\frac{X^i(t)-(X_*)^i(t-L,t)}{(X^*)^i(t-L,t)-(X_*)^i(t-L,t)}}_{i\text{-th position}},\ldots,0\right). \tag{3.9}$$

Conversely, the **anti-momentum (or mean-reversion) strategy** assumes assets far from their highs are undervalued and due for a rebound. Its weight is proportional to the distance from the peak:

$$\theta_{\text{anti}}^{(i)}(t) = \left(0,\ldots,\underbrace{\frac{(X^*)^i(t-L,t)-X^i(t)}{(X^*)^i(t-L,t)-(X_*)^i(t-L,t)}}_{i\text{-th position}},\ldots,0\right). \tag{3.10}$$

The vectors for each single-asset strategy are valid portfolio weights as they are non-negative and sum to one. These m individual strategies are first aggregated into a unified momentum portfolio and a unified anti-momentum portfolio using the capital-weighted aggregation method (Equation 3.11). These two meta-strategies are then further aggregated into a final combined momentum/anti-momentum strategy. This hierarchical aggregation creates a complex, adaptive strategy that dynamically balances between trend-following and contrarian impulses based on their recent performance.

3.3 Strategy Aggregation

The project explores the aggregation of multiple individual strategies into a meta-strategy. The idea is to combine the strengths of individual allocation rules while reducing their weaknesses. If K strategies with portfolio values $V_t^{(k)}$ and weight vectors $\boldsymbol{\theta}_t^{(k)}$ are to be combined, the aggregated strategy $\widehat{\boldsymbol{\theta}}_t$ is formed by a capital-weighted average:

$$\widehat{\boldsymbol{\theta}}_t = \frac{\sum_{k=1}^{K} \boldsymbol{\theta}_t^{(k)} \cdot V_t^{(k)}}{\sum_{k=1}^{K} V_t^{(k)}}. \tag{3.11}$$

The value of this new aggregated portfolio is simply the average of the constituent values:

$$\widehat{V}_t = \frac{1}{K} \sum_{k=1}^{K} V_t^{(k)}. \tag{3.12}$$

This mechanism allows for the creation of hierarchical portfolios, such as an aggregate of BAH, CRP, momentum, and anti-momentum, which is then treated as a standalone strategy for further analysis and benchmarking.

3.4 Modeling Transaction Costs

A realistic evaluation of portfolio strategies must incorporate the frictions associated with trading. In financial markets, these costs arise from multiple sources, including bid–ask spreads, brokerage fees, and market impact. Even when simplified, transaction costs can have a significant cumulative effect, particularly for high-turnover strategies.

A cost is applied to every trade in each asset. The change in capital allocated to asset i at time t is calculated as:

$$\Delta_t^i = \theta_t^i V_t - \theta_{t-1}^i V_{t-1} \frac{X_t^i}{X_{t-1}^i}. \tag{3.13}$$

If $\Delta_t^i > 0$ the position in asset i is increased; if $\Delta_t^i < 0$, it is reduced. The transaction cost is the maximum of two terms: a proportional fee equal to (0.1%) of the trade's absolute value or a fixed minimum fee of one currency unit. Costs apply only if a trade occurs ($|\Delta_t^i| > 0$). The cost at time t is therefore:

$$C_t = \sum_{i=1}^{m} \left(0.001 \cdot |\Delta_t^i| \vee 1\right) \cdot \mathbb{I}_{\{|\Delta_t^i|>0\}}. \tag{3.14}$$

Cumulative costs over the investment horizon are then $C_T(\boldsymbol{\theta}) = \sum_{t=1}^{T} C_t$, so that the net terminal wealth is:

$$V_T^{\text{net}}(\boldsymbol{\theta}) = V_T(\boldsymbol{\theta}) - C_T(\boldsymbol{\theta}). \tag{3.15}$$

This cost structure explicitly penalizes high-turnover strategies like CRP and momentum, while the BAH strategy, after the initial purchase, incurs no costs. Without this adjustment, results risk overstating performance and misleading conclusions about relative efficiency.

3.5 Benchmarking and Performance Metrics

The performance of all strategies is evaluated against a passive market index benchmark. This benchmark is constructed as an equi-weighted portfolio of the assets at their initial weights, with no subsequent rebalancing or cost deductions:

$$\text{Index}_t = \left(\frac{\sum_{i=1}^{m} X_t^i}{\sum_{i=1}^{m} X_0^i} \right) \cdot V_0. \tag{3.16}$$

This provides a neutral baseline reflecting the general price movement of the entire asset universe.

The primary metric for comparing strategy performance is the **logarithmic portfolio value** over time, which intuitively shows compounded growth and periods of drawdown. Furthermore, the **Maximum Drawdown (MDD)** is calculated for each strategy. The MDD is defined as the maximum peak-to-trough decline in the portfolio's value before a new peak is achieved, measured as:

$$\text{MDD} = \max_{\tau \in (0,T)} \left[\frac{\max_{s \in (0,\tau)} V_s - V_\tau}{\max_{s \in (0,\tau)} V_s} \right]. \tag{3.17}$$

This metric is a crucial gauge of strategy risk and investor pain, as it captures the worst-case historical loss.

3.6 Empirical Implementation and Data Processing

The models are implemented in Python using standard scientific libraries: NumPy for numerical operations, Pandas for data handling, Matplotlib for visualization, and yFinance for price data. Data consist of adjusted daily closing prices over a 10-year horizon, obtained via the Yahoo Finance API. Adjusted prices are used because they account for dividends and stock splits, giving an accurate measure of total returns. To reduce daily noise while avoiding the coarseness of monthly data, the series are resampled to weekly frequency, which smooths short-term volatility but preserves medium-term trend reversals relevant for momentum strategies.

Two asset universes are analyzed to test robustness. The first, the *"Magnificent Seven"*, includes leading technology stocks (`AAPL`, `MSFT`, `NVDA`, `AMZN`, `META`, `GOOGL`, `TSLA`). The second, the *"Weakest Seven"*, includes underperforming S&P 500 firms (`WBD`, `PARA`, `APA`, `WBA`, `PCG`, `VTRS`, `DOW`), representing adverse market conditions.

All portfolios are initialized with a nominal value $V_0 = 10,000$. Each asset's price series is normalized by its initial price X_0^i to convert absolute prices into cumulative growth factors, ensuring comparability across assets with different price scales. Weekly returns R_t^i are computed from these normalized series. Finally, the data are cleaned by removing missing values to maintain alignment across all assets in each universe.

3.7 Numerical Robustness and Implementation Safeguards

The simulation code incorporates several safeguards to ensure numerical stability and robustness. These include explicit checks to prevent division by zero during normalization

and score calculations, the handling of NaN values that may arise during the initial rolling window period, and an initialization protocol where momentum-based strategies remain flat (100% cash) until the 52-week lookback window is fully populated with data. This prevents the generation of spurious signals from insufficient data. Finally, all portfolio values are plotted in logarithmic scale, which not only linearizes compounded returns but also facilitates comparison across portfolios of different magnitudes. This choice avoids distortion from exponential growth effects and is consistent with common practice in financial econometrics. These adjustments were further supported by preliminary experiments conducted in this project, which indicated that weekly sampling, an annual lookback horizon, and log-scale visualization provided the clearest and most representative insights into the strategies under study.

Chapter 4

Results

This section reports the empirical outcomes for four portfolio strategies: Buy-and-Hold (BAH), Constant Rebalanced Portfolio (CRP), a combined Momentum and Anti-Momentum strategy (M+AM), and an Aggregated strategy incorporating all three. Performance is assessed across two contrasting asset universes, using compounded growth (shown on a logarithmic scale) and Maximum Drawdown (MDD) as key metrics. The results highlight how strategy performance depends strongly on market regime and demonstrate the stabilizing effect of diversification.

4.1 Performance in a High-Growth Regime: The "Magnificent Seven"

Figure 4.1 shows the performance of all strategies applied to the high-growth "Magnificent Seven" universe. The logarithmic scale illustrates compounded growth and relative drawdowns.

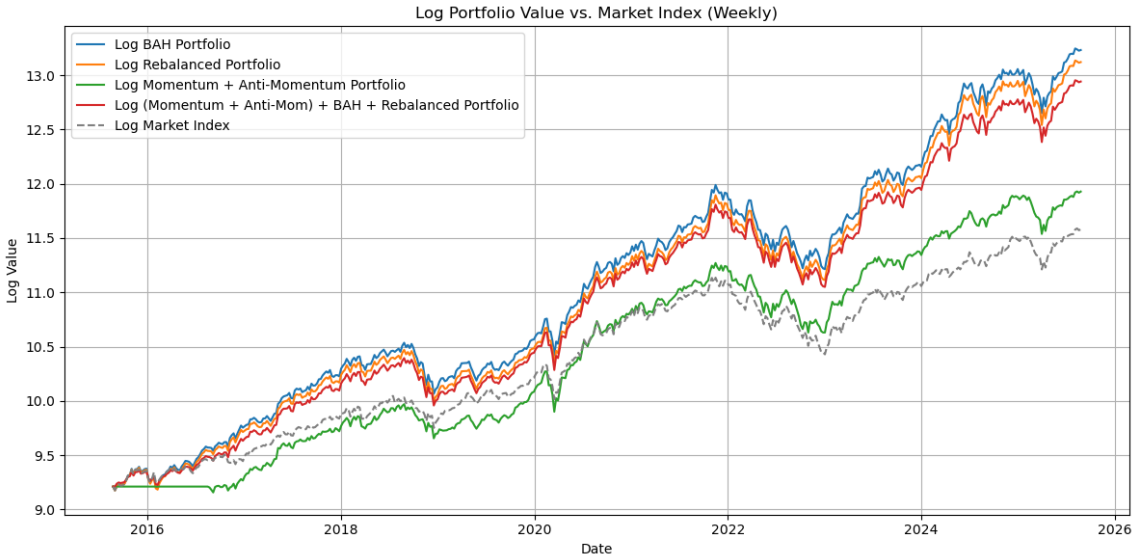


Figure 4.1: Logarithmic portfolio value evolution for the "Magnificent Seven" universe (2015–2025). The Aggregated strategy achieves the highest terminal value and a smoother growth path, while the Momentum + Anti-Momentum strategy underperforms due to turnover costs.

The Buy-and-Hold (BAH) strategy provided a clear benchmark, benefiting from the strong upward trend in technology stocks over the decade. However, it was fully exposed to market shocks, with pronounced declines during the COVID-19 crash of 2020 and the inflation-driven selloff of 2022. The Constant Rebalanced Portfolio (CRP) reached a similar terminal value, though through a different path. Its rebalancing mechanism provided limited gains during volatile periods but imposed costs in extended rallies, preventing it from outperforming BAH.

The Momentum + Anti-Momentum (M+AM) strategy underperformed both benchmarks in this universe. The combination of high volatility, frequent trend reversals, and elevated turnover led to poor timing and significant transaction costs, which eroded returns compared to BAH.

The Aggregated strategy delivered the strongest outcome, achieving both the highest terminal value and the smoothest equity curve. By reallocating across BAH, CRP, and M+AM based on recent performance, it adjusted exposure to prevailing conditions and reduced drawdowns. This result supports Cover's universal portfolio theory, showing that while no single strategy dominates, a diversified ensemble can adapt effectively to shifting market regimes.

Table 4.1: Maximum Drawdown (MDD) for the "Magnificent Seven" Universe

Strategy	**Max Drawdown**
Buy-and-Hold (BAH)	-54.18%
Constant Rebalanced Portfolio (CRP)	-54.45%
Momentum + Anti-Momentum (M+AM)	-47.32%
Aggregated Strategy	**-43.15%**
Market Index	-50.64%

The risk analysis, summarized by Maximum Drawdown in Table 4.1, supports the visual evidence. Both BAH and CRP experienced deep drawdowns above 54%, reflecting their high exposure to market-wide risk. The M+AM strategy, while weaker in absolute returns, reduced drawdowns by about 7 percentage points, as its momentum component tended to shift away from assets entering steep declines. The Aggregated strategy delivered the strongest risk-adjusted outcome, lowering peak-to-trough losses by more than 11 percentage points relative to BAH. It also outperformed the passive market index by 7.5 percentage points, highlighting its effectiveness as a risk-management approach.

4.2 Stress Testing in a Secular Decline: The "Weakest Seven"

Figure 4.2 shows the performance of all strategies in the "Weakest Seven" universe. This setting, characterized by persistent declines, provides a strong test of robustness.

The BAH strategy produced a steady and substantial loss due to its static exposure. The CRP strategy performed even worse. Its rebalancing mechanism—buying assets that had fallen the most—amplified losses in the absence of mean reversion, leading to the weakest outcome with a drawdown above 72%.

The Momentum + Anti-Momentum (M+AM) strategy was the best performer in this environment. The momentum rule systematically avoided the worst performers, while the

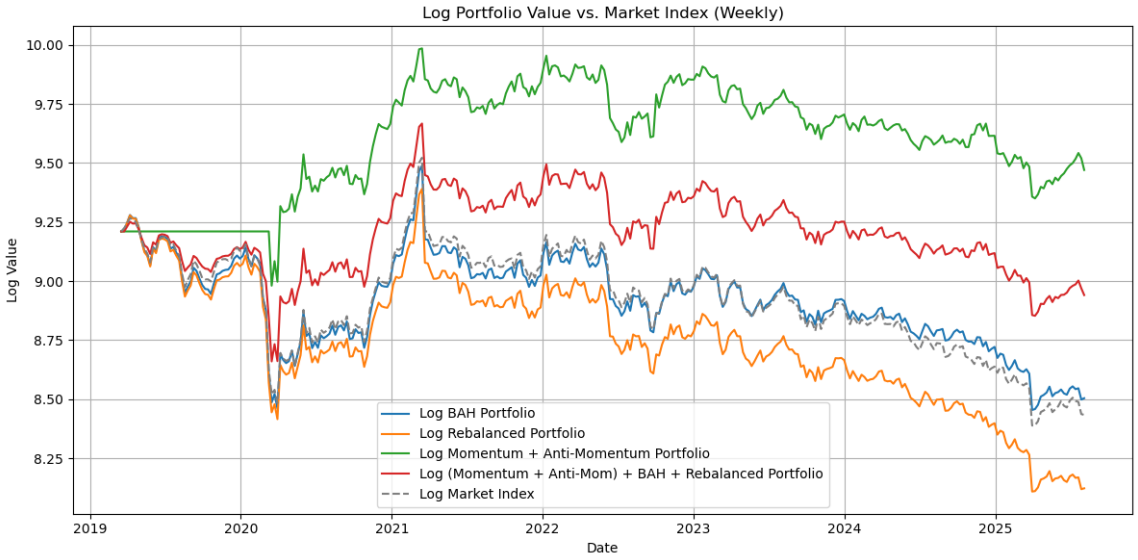


Figure 4.2: Logarithmic portfolio value evolution for the "Weakest Seven" universe. The Constant Rebalanced Portfolio (CRP) fails catastrophically. The Momentum + Anti-Momentum and Aggregated strategies significantly outperform by avoiding the worst-performing assets.

anti-momentum rule struggled with assets that continued to fall. Overall, the momentum signal dominated, allowing the strategy to limit losses relative to BAH and CRP. This highlights the effectiveness of trend-following approaches in prolonged bear markets.

The Aggregated strategy also showed resilience. While it could not avoid losses entirely, it outperformed BAH and CRP by allocating away from the worst-performing CRP and toward M+AM. Its drawdown was nearly identical to M+AM, but its main advantage lies in its adaptability: it performs consistently across both bull and bear regimes without requiring prior knowledge of which strategy will dominate.

Table 4.2: Maximum Drawdown (MDD) for the "Weakest Seven" Universe

Strategy	**Max Drawdown**
Buy-and-Hold (BAH)	-64.48%
Constant Rebalanced Portfolio (CRP)	-72.25%
Momentum + Anti-Momentum (M+AM)	-47.03%
Aggregated Strategy	**-43.00%**
Market Index	-67.84%

The Maximum Drawdown results in Table 4.2 confirm the poor performance of CRP and the relative strength of adaptive strategies. The Aggregated strategy recorded a drawdown of 43%, nearly identical to its 43.15% drawdown in the high-growth universe, indicating consistent risk control across regimes. Relative to the market index, it reduced drawdowns by about 25 percentage points, underscoring its effectiveness in limiting capital losses.

4.3 Visual Analysis of Portfolio Composition

Further insight can be drawn from Figure 4.3, which decomposes the evolution of the BAH and M+AM strategies for the "Magnificent Seven" universe. The upper panel, showing normalized asset prices (X_t/X_0), highlights the exceptional growth of a few dominant firms—most notably NVDA—relative to their peers. This divergence drives much of the portfolio dynamics observed in the subsequent plots.

The middle panel shows the implicit market-capitalization-like weights of the BAH portfolio. As NVDA's price appreciation outpaced all others, its portfolio weight expanded steadily to nearly 70% of the portfolio, while the relative shares of the other constituents shrank. This illustrates the concentrated exposure risk of BAH: while it benefits from compounding in the best-performing assets, it leaves the portfolio highly vulnerable to reversals in those same names.

The bottom panel presents the allocation dynamics of the M+AM strategy. Unlike BAH, it actively redistributes capital across assets, with weights shifting frequently as momentum and anti-momentum signals evolve. Periods of heightened volatility, such as 2022–2023, show sharp reallocations and even temporary concentration in single names when they exhibited relative strength. While this active rebalancing introduces higher turnover and cost, it also demonstrates the strategy's flexibility, enabling it to capture emerging winners and reduce exposure to temporary laggards. Overall, this visualization highlights the structural difference between passive concentration in BAH and the adaptive, signal-driven reallocations in M+AM.

Figure 4.4 shows the evolution of the BAH and M+AM strategies for the "Weakest Seven" universe. The upper plot, showing normalized asset prices (X_t/X_0), visually confirms the secular decline of all assets, with some (e.g., PARA) effectively trending to zero. The middle plot reveals the dynamic, market-capitalization-like weights of the BAH portfolio. As certain assets crash more severely than others, their portfolio weight automatically decreases, which is the only defensive mechanism available to this strategy.

The bottom plot, showing the weights of the M+AM strategy, is arguably the most revealing. It shows a highly dynamic and active allocation. The strategy aggressively shifts capital, at times concentrating heavily in just one or two assets that it deems strongest (least weak) according to its signals, while simultaneously avoiding the precipitous decliners. This active risk-taking and concentration is what allowed it to sidestep the worst of the downturn, a feat impossible for the passive strategies. This visualization underscores the active decision-making that drives the performance differential.

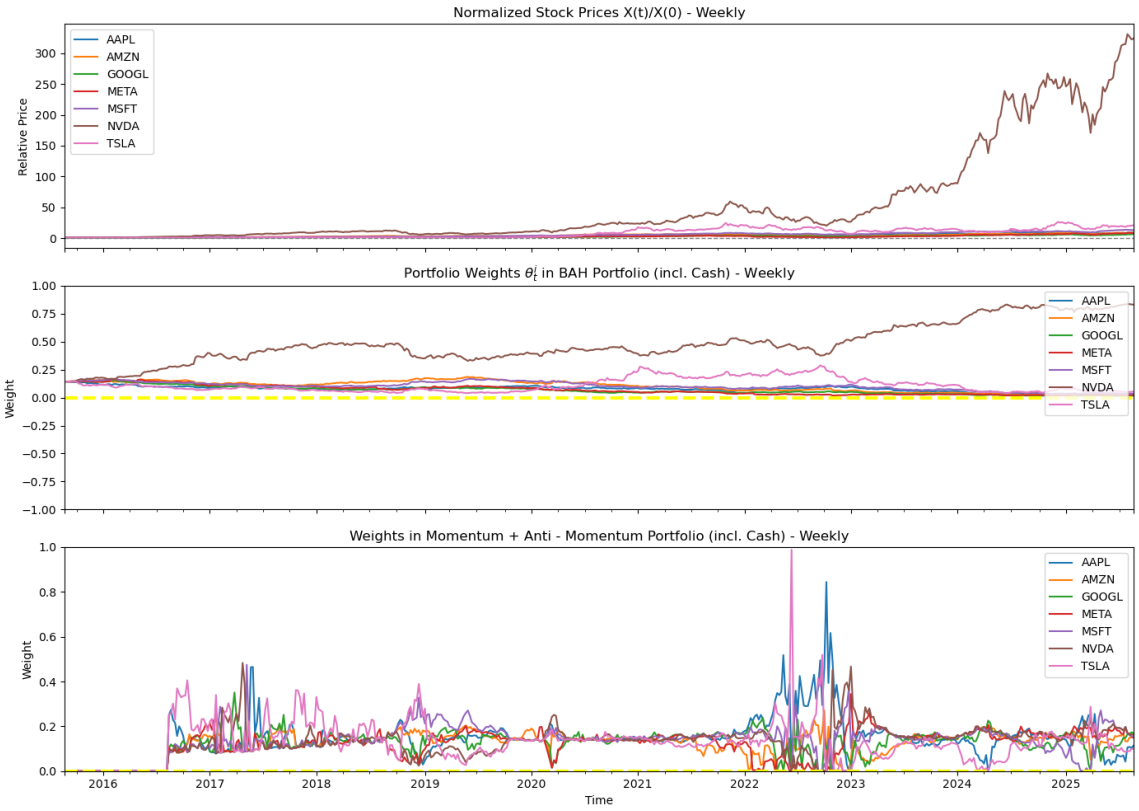


Figure 4.3: Evolution of BAH and M+AM strategies in the "Magnificent Seven" universe. Top: Normalized asset prices show diverging performance, with NVDA driving exceptional cumulative gains. Middle: BAH portfolio weights concentrate on top performers, illustrating high exposure to individual winners. Bottom: M+AM dynamically reallocates based on signals, adapting to changing conditions but incurring turnover costs.

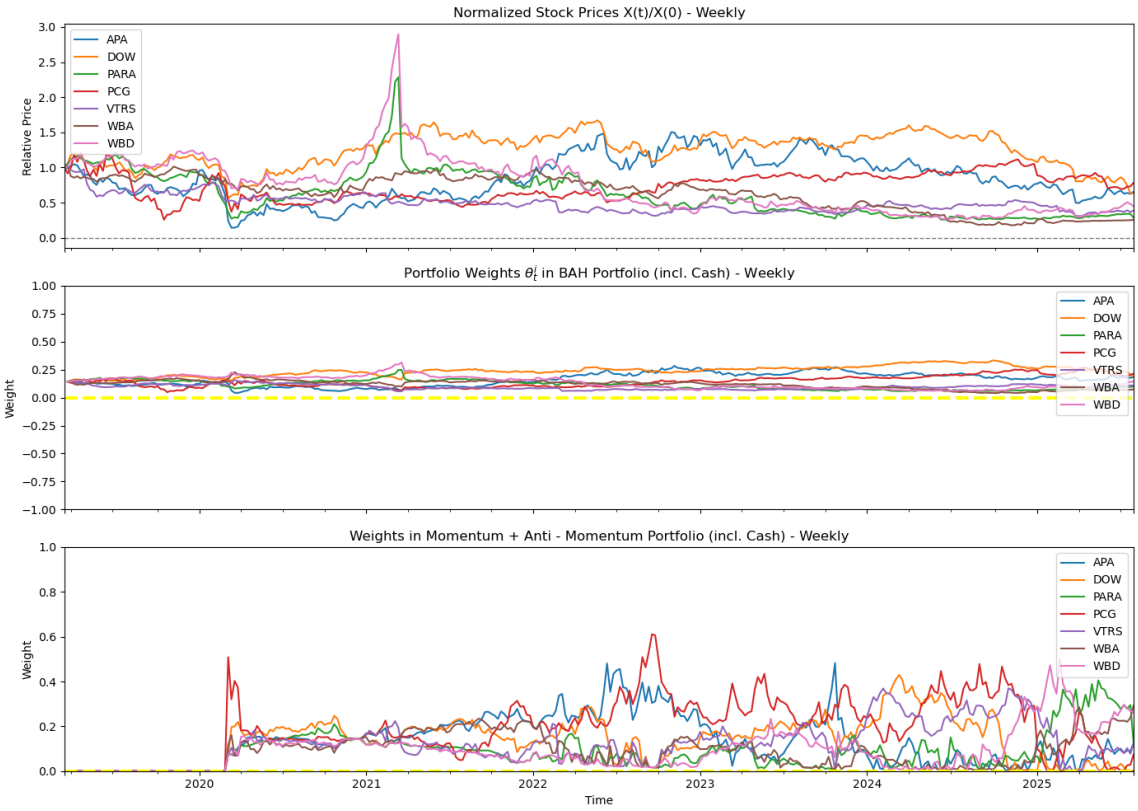


Figure 4.4: Portfolio composition analysis for the "Weakest Seven". Top: Normalized asset prices show universal decline. Middle: The evolving weights of the BAH strategy. Bottom: The active, concentrated allocations of the Momentum + Anti-Momentum strategy, which allowed for effective capital preservation.

Chapter 5

Discussion and Conclusion

The empirical investigation undertaken in this thesis evaluates five fundamental and hybrid portfolio allocation strategies—Buy and Hold (BAH), Constant Rebalanced Portfolio (CRP), Momentum, Anti-Momentum (Contrarian), and Aggregated strategies—across two sharply contrasting market environments: the high-growth "Magnificent Seven" and a set of underperforming, structurally impaired equities. This dual-market analysis allowed for both stress testing and performance benchmarking across typical and adverse financial regimes. The principal performance metric applied throughout this study was Maximal Drawdown (MDD)—a decision informed by the real-world emphasis on downside risk rather than volatility. While traditional performance measures such as Sharpe ratio or compound annual growth rate (CAGR) dominate academic and professional finance, this thesis posits that path-dependent losses, particularly the depth and duration of capital erosion, are more aligned with investor sensitivity and institutional mandates for capital preservation. By prioritizing drawdown analysis, we aim to address the often underemphasized yet profoundly relevant question of how much pain an investor must tolerate along the path to long-term gains.

The results strongly reinforce the well-established idea that no single portfolio strategy is universally optimal. In the context of the Magnificent Seven, Buy and Hold performed competitively, owing to the strong and sustained uptrends in firms like NVDA, AAPL, and MSFT. These stocks benefited from both technological tailwinds and broad investor optimism, which supported the Efficient Market Hypothesis assumption of secular growth—a foundational tenet of the BAH strategy (Fama, 1970). However, even in this favorable environment, BAH incurred a drawdown exceeding 50

Momentum-based strategies fared slightly better, achieving lower drawdown than both BAH and CRP. By overweighting recent winners and underweighting losers, momentum captured the essence of behavioral finance insights—such as underreaction and investor herding (Jegadeesh and Titman, 1993; Hong and Stein, 1999). Still, momentum's performance was constrained by the cost of turnover and susceptibility to regime changes. Perhaps most instructive is the outperformance of the Aggregated strategy, which dynamically combined the above methods. By allocating capital in proportion to sub-strategy performance, the aggregator maintained exposure to the most successful behaviours while dampening the risk associated with any single method. This approach not only produced the highest terminal value but also the lowest MDD, demonstrating robustness both in compounding and in capital protection.

In the stress-testing scenario, where portfolios were applied to weak or declining stocks, the distinctions between strategies became more pronounced. The Buy and

Hold approach failed predictably, reflecting its structural inflexibility and sensitivity to poor asset selection. CRP exacerbated losses due to its insistence on reallocating into declining assets—ironically increasing exposure to value traps and downward momentum. Contrastingly, Momentum once again demonstrated defensive characteristics, reducing drawdowns significantly. Its dynamic disengagement from underperformers provided a form of implicit risk control, allowing the strategy to avoid deep drawdowns despite operating in a broadly declining market. The Aggregated strategy again offered superior protection. It succeeded in preserving capital relative to the market and other strategies, despite being exposed to a universe devoid of consistent winners. This highlights the key insight of the thesis: diversification across strategic logic—rather than simply across assets—yields resilience in environments where traditional diversification fails.

Moreover, the combination of momentum and anti-momentum in the aggregator enabled the portfolio to toggle adaptively between trend capture and reversion-seeking, a balance that individual strategies could not maintain. The key advantage here lies in the time-varying nature of market regimes. Neither trending nor mean-reverting behaviour dominates persistently; thus, static allocation rules inevitably underperform in some settings. Strategy aggregation provides a path toward meta-adaptivity, allocating resources not just to assets, but to decision paradigms that evolve in response to market structure.

Beyond empirical outcomes, this thesis contributes methodologically in several areas. Each strategy is articulated in terms of formal portfolio mappings, ensuring analytical clarity and consistency. The implementation includes realistic frictions via proportional fees and minimum execution costs, reflecting institutional trading conditions. The weekly rebalancing horizon avoids overfitting to high-frequency noise, and rolling-window logic in momentum strategies aligns with academic precedent. Furthermore, the modular aggregation framework provides reusable simulation tools for multi-level portfolio nesting. These implementations form a foundation that can be extended to incorporate machine learning-based meta-strategies, regime-switching detection, or conditional rebalancing heuristics.

The consistent emphasis on Maximal Drawdown distinguishes this thesis from many traditional finance studies. Volatility-based risk metrics—such as standard deviation or Value at Risk—fail to account for investor psychology during periods of deep capital loss. MDD, in contrast, speaks directly to the emotional and institutional thresholds of loss tolerance. This focus is especially relevant in light of recent market shocks, including the COVID-19 crash and regional banking crises, where temporary value erosion caused major outflows even from fundamentally sound portfolios. Institutional mandates often incorporate drawdown-based constraints; therefore, a strategy that minimizes MDD—even at the cost of slightly lower CAGR—may be more implementable and more acceptable in real-world investment settings.

Naturally, this empirical study has limitations. It makes several simplifying assumptions, including a static universe of assets known ex ante, and it does not model liquidity, bid-ask spreads, taxes, or slippage. Additionally, momentum and anti-momentum signals are computed purely from price levels, excluding volume, news, or fundamentals. Furthermore, while strategy aggregation proved successful, the proportional capital weighting scheme used is just one among many. Future research could explore Bayesian updating, reinforcement learning, or drawdown-conditioned allocations. The integration of fundamental screening into the contrarian strategy would also enhance robustness against value traps. Finally, it would be valuable to validate these findings on broader universes,

including emerging markets, small caps, or sector-rotated portfolios.

In conclusion, this thesis demonstrates that portfolio strategy design is most robust when it incorporates multiple market hypotheses and adapts to shifting conditions. By analyzing classical and behavioural strategies under both bullish and bearish asset universes, we confirm the inadequacy of static methods and the practical strength of strategic diversification. Aggregated portfolios, when constructed carefully, outperform not only in returns but in risk-resilience, especially as measured by Maximal Drawdown. The practical implication is clear: diversify not just across assets, but across beliefs. Strategies that are overcommitted to one market worldview—be it efficiency, mean reversion, or trend persistence—are prone to fragility. Strategy aggregation, framed and evaluated under drawdown risk, represents a promising path forward for investors seeking long-term growth without unacceptable capital erosion.